

MATH 230-2 Worksheet 1

Tuesday, April 8th

Problem 1. Consider the solid that lies above the square $R = [0, 2] \times [0, 2]$ and below the paraboloid $z = 16 - x^2 - 2y^2$.

- (a) Estimate the volume of the solid using a Riemann sum: divide R into four equal squares and choose sample points to be in the upper right corner of each square.
- (b) Calculate the true volume of the solid using a double integral.

Problem 2. Let D be the unit disk.

- (a) Show that

$$\iint_D x \, dA = \iint_D y \, dA = 0.$$

- (b) Find the volume of the region above the unit disk D in the xy -plane and below the plane $z = 16 - x - y$.

Problem 3. For each part, sketch the region D indicated. Then write (but do not evaluate) the integral

$$\iint_D f(x, y) \, dA$$

as an iterated integral in both possible orders.

- (a) D is the region in the first quadrant bounded by the curves $y = x$ and $y = x^3$.
- (b) D is the region in the first quadrant bounded by the curves $x = y^2$ and $x = 4$.

Problem 4. Let D be the triangle formed by the vertices $(0, 0)$, $(1, 0)$, and $(1, 1)$. Calculate

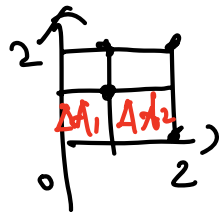
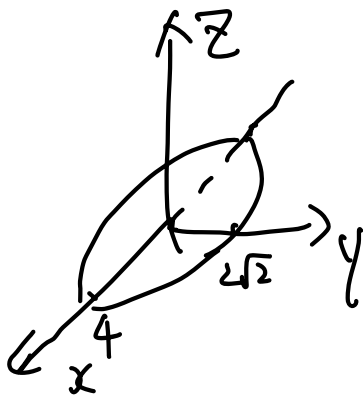
$$\iint_D e^{x^2} \, dA.$$

(Hint: one order of integration will be better than the other!)

Problem 5. Evaluate the following integral by reversing the order of integration:

$$\int_0^1 \int_{\arcsin y}^{\pi/2} 2y \cos x \, dx \, dy.$$

Problem 1



$$(a) \quad V = \iint_R f(x,y) dA$$

$$= \iint_R (16 - x^2 - 2y^2) dA$$

$$\approx f(x_1, y_1) \Delta A_1$$

$$+ f(x_2, y_2) \Delta A_2$$

$$+ f(x_3, y_3) \Delta A_3$$

$$+ f(x_4, y_4) \Delta A_4$$

$$= f(1, 1) \cdot 1 + f(2, 1) \cdot 1 + f(1, 2) \cdot 1$$

$$+ f(2, 2) \cdot 1$$

$$= (16 - 3) + (16 - 4 - 2) + (16 - 1 - 8) + (16 - 4 - 8)$$

$$= 13 + 10 + 7 + 4 = 34$$

$$(b) \quad \int_0^2 \int_0^2 (16 - x^2 - 2y^2) dx dy$$

$$= \int_0^2 \left(16x - \frac{1}{3}x^3 - 2y^2x \right) \Big|_0^2 dy$$

$$= \int_0^2 32 - \frac{1}{3} \cdot 2^3 - 2y^2 \cdot 2 dy$$

$$= 32y - \frac{1}{3} \cdot 2^3 \cdot y - 2 \cdot \frac{1}{3} y^3 \cdot 2 \Big|_0^2$$

$$= 64 - \frac{16}{3} - \frac{32}{3} = 64 - \frac{48}{3} = 48$$

Problem 2

(a) By polar coordinates transform

$$\iint_D x \, dA = \int_0^{2\pi} \int_0^1 r \cos \theta \cdot r \, dr \, d\theta = 0$$

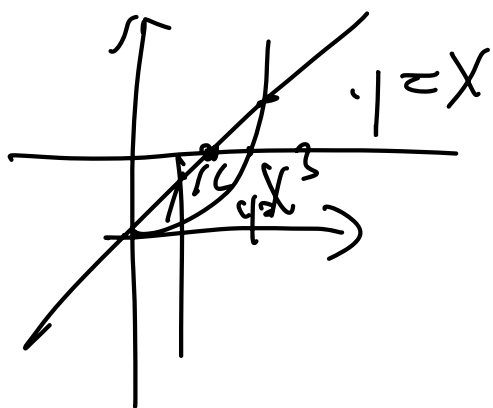
Similarly for $\iint_D y \, dA = 0$

Or one may argue by symmetry.

$$\begin{aligned} (b) \iint_D (16 - x - y) \, dA &= \iint_D 16 \, dA - \iint_D x \, dA - \iint_D y \, dA \\ &\stackrel{\text{by (a)}}{=} \iint_D 16 \, dA - 0 - 0 \\ &= 16 \cdot \text{Area}(D) = 16 \cdot \pi. \end{aligned}$$

Problem 3

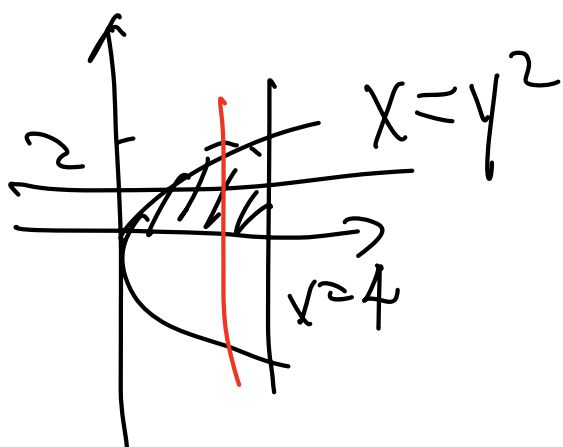
(a)



$$\int_0^1 \int_{x^3}^x f(x,y) dx dy$$

$$\text{or } \int_0^1 \int_{x^3}^x f(x,y) dy dx$$

(b)



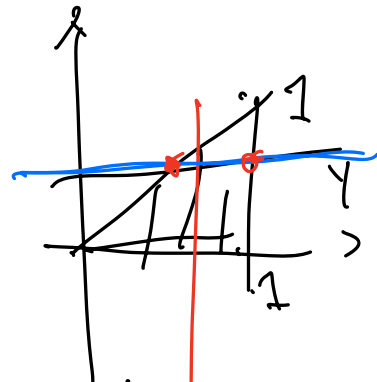
$$\int_0^2 \int_{y^2}^4 f dx dy$$

$$\text{or } \int_0^4 \int_0^{\sqrt{x}} f dy dx$$

Problem 4

$\int_0^1 \int_0^1 e^{x^2} dx dy$ is hard to solve!

So we compute, instead,



$$\int_0^1 \int_0^x e^{x^2} dy dx = \int_0^1 x e^{x^2} dx$$

$$= \frac{1}{2} e^{x^2} \Big|_0^1 = \frac{1}{2} (e - 1)$$

Problem 5

$$\int_0^1 \int_0^{\arcsin y} 2y \cos x dx dy$$

$$= \int_0^{\pi/2} \int_0^{\sin x} 2y \cos x dy dx$$

$$= \frac{1}{3} (\sin x)^3 \Big|_0^{\pi/2} = \frac{1}{3}$$

